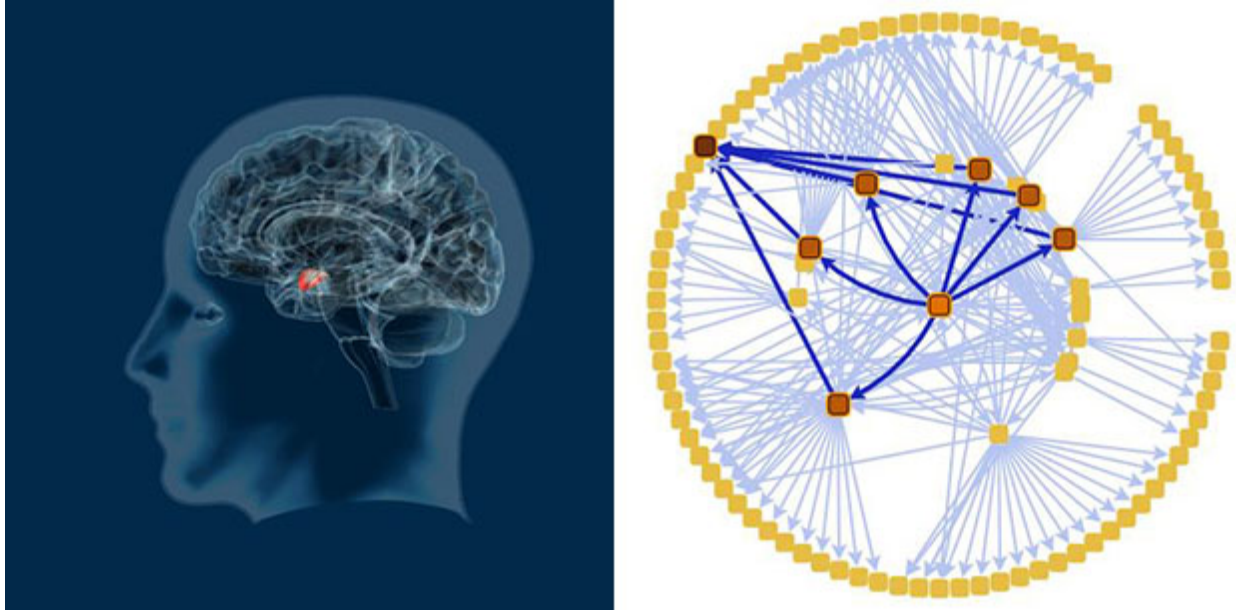


Brain, Neurons, Cognition: Computational Neuroscience

 blog.wolfram.com/2017/06/06/brain-neurons-cognition-computational-neuroscience

June 6, 2017



As the next phase of Wolfram Research's endeavor to make biology computable, we are happy to announce the recent release of neuroscience-related content.

The most central part of the human nervous system is the brain. It contains roughly 100 billion neurons that act together to process information, subdivided functionally and structurally into areas specialized for certain tasks. The brain's anatomy, the characteristics of neurons and cognitive maps are used to represent some key aspects of the functional organization and processing abilities of our nervous system. Our new neuroscience content will give you a sneak peek into the amazing world of neuroscience with some facts about brains, neurons and cognition.

Brain Anatomy and Network

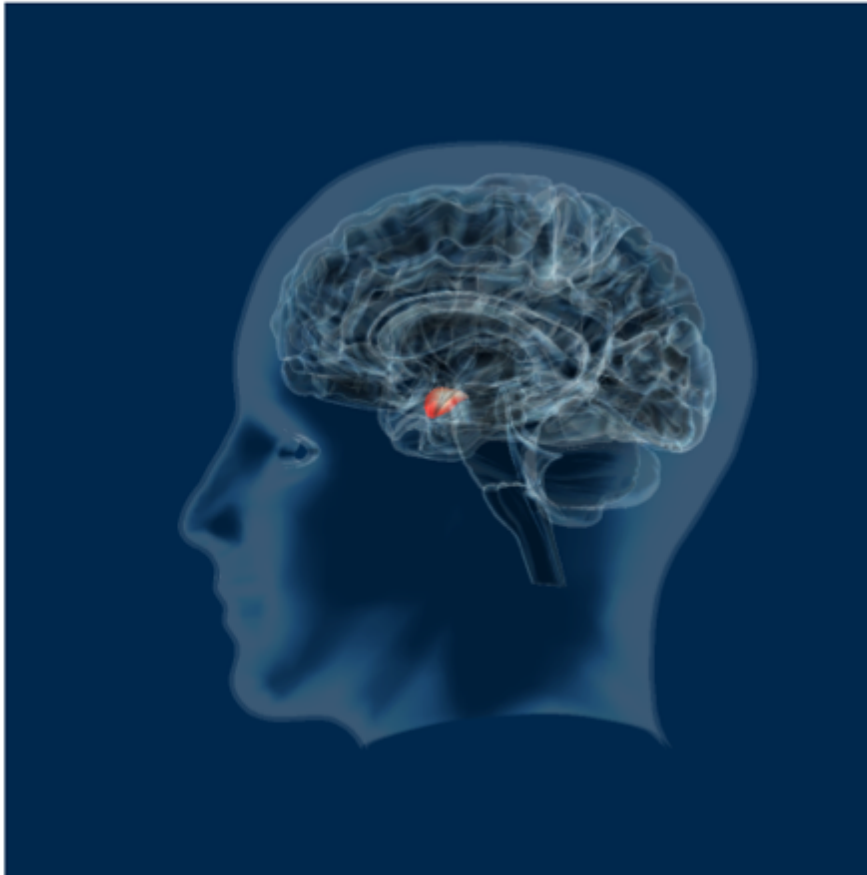
A primal part of the brain, the amygdala, is the well-studied cognitive area responsible for the emotional process circuitry and has active roles in emotional state, memory, face recognition and decision making. The amygdala is located near the brainstem, close to the center of the brain and, as its name suggests, is shaped like an almond:

```

In[1]:= light = Join[{"Ambient", Black},
  Table[{"Directional", Hue[.58, .5, 1], ImageScaled[{Sin[x], Cos[x], -.5}], Pi/2},
    {x, 0, 2 Pi - 2 Pi/8, 2 Pi/8}]];
AnatomyPlot3D[
  {AnatomyForm[<|_ → Directive[Specularity[White, 30], Hue[.58, 0, 1, .12], Lighting → light],
    amygdala (anatomical structure) → Glow[Red]>], amygdala (anatomical structure),
    brain (anatomical structure), skin (anatomical structure)], Background → Hue[.58, 1, .3],
  ViewPoint → Right, SphericalRegion → True,
  PlotRange → {{Automatic, 0}, Automatic, {1400, Automatic}},
  BaseStyle → {RenderingOptions → {"DepthPeelingLayers" → 20}}, ImageSize → 500]

```

Out[2]=



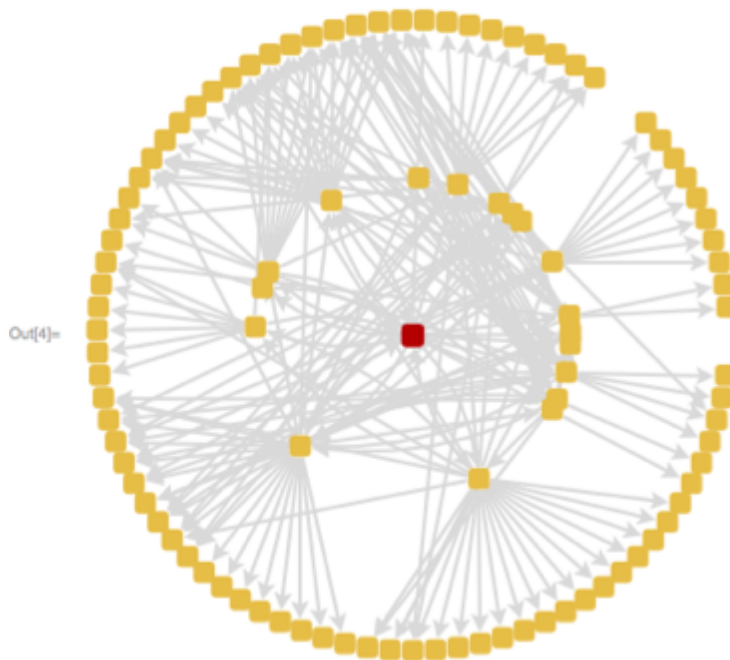
Outgoing connections from the amygdala can be found with the "NeuronalOutput" property:

```
In[3]:= output = amygdala (anatomical structure) ["NeuronalOutput"]
```

```
Out[3]= { lateral hypothalamic region , locus ceruleus , prefrontal cortex ,  
          central amygdaloid nucleus , medial preoptic nucleus , ventral tegmental area ,  
          nucleus subceruleus , lateral parabrachial nucleus , medial parabrachial nucleus ,  
          solitary tract nuclear complex , premotor cortex , central gray substance of midbrain ,  
          dorsal nucleus of vagus nerve , subparabrachial nucleus , cuneiform nucleus ,  
          anterior hypothalamic region , orbital cortex , primary motor cortex }
```

Here we see a visualization of the output connectivity of the amygdala in two layers:

```
In[4]:= ag =  
NestGraph[  
  Cases[EntityValue[#, EntityProperty["AnatomicalStructure", "NeuronalOutput"]], _Entity] &,  
  amygdala (anatomical structure) , 2, VertexSize -> 2, EdgeStyle -> LightGray,  
  GraphStyle -> "SmallNetwork", GraphLayout -> "RadialEmbedding",  
  VertexLabels -> Placed["Name", Tooltip], GraphHighlight -> amygdala (anatomical structure) ,  
  ImageSize -> 400]
```

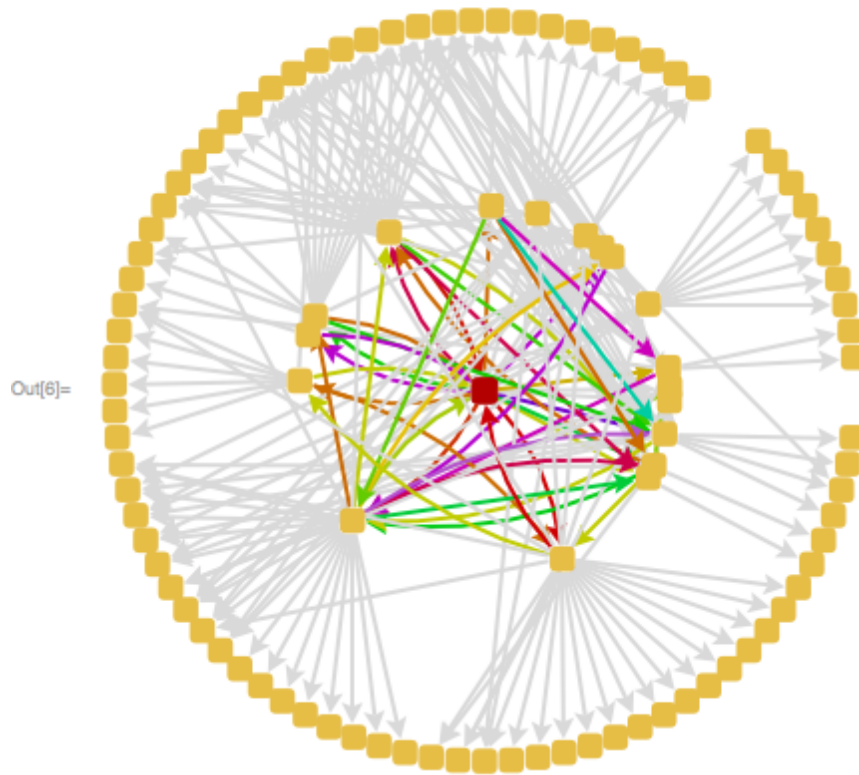


Just as in a simple network, we could do additional computations on other networks. Like many other biological systems, our nervous system is hardwired to receive positive and negative feedback. Feedback is one of the key aspects of the brain's information processing; it allows the augmentation or decrease of the efficacy of transmission, as well as fine-tuning for the resulting outputs.

Find the loop and highlight in the above graph:

```
In[5]:= cycle = FindCycle[ag, Infinity, All];
```

```
In[6]:= HighlightGraph[ag, cycle]
```



Or find the specific circuit that comprises the combination amygdala-prefrontal cortex. The prefrontal cortex has the primary role in decision making, and therefore the amygdala-prefrontal cortex connectivity plays an essential role in modulating responses to emotional experiences:

```
In[7]:= circuit = FindCycle[{ag, Entity["AnatomicalStructure", "PrefrontalCortex"]}]
```

Out[7]= { { prefrontal cortex → amygdala , amygdala → anterior hypothalamic region ,
 anterior hypothalamic region → central gray substance of midbrain ,
 central gray substance of midbrain → solitary tract nuclear complex ,
 solitary tract nuclear complex → central amygdaloid nucleus ,
 central amygdaloid nucleus → ventral tegmental area ,
 ventral tegmental area → prefrontal cortex } }

```

In[8]:= HighlightGraph[ag, Join[circuit, VertexList[Graph[First@circuit]],
{Entity["AnatomicalStructure", "AnteriorCingulateGyrus"]}], GraphHighlightStyle -> "Thick"]

```



We can also identify the minimum-cost flow between the amygdala and the spinal cord. The spinal cord processes signals from the brain and transmits them to other parts of the body to excite motor response:

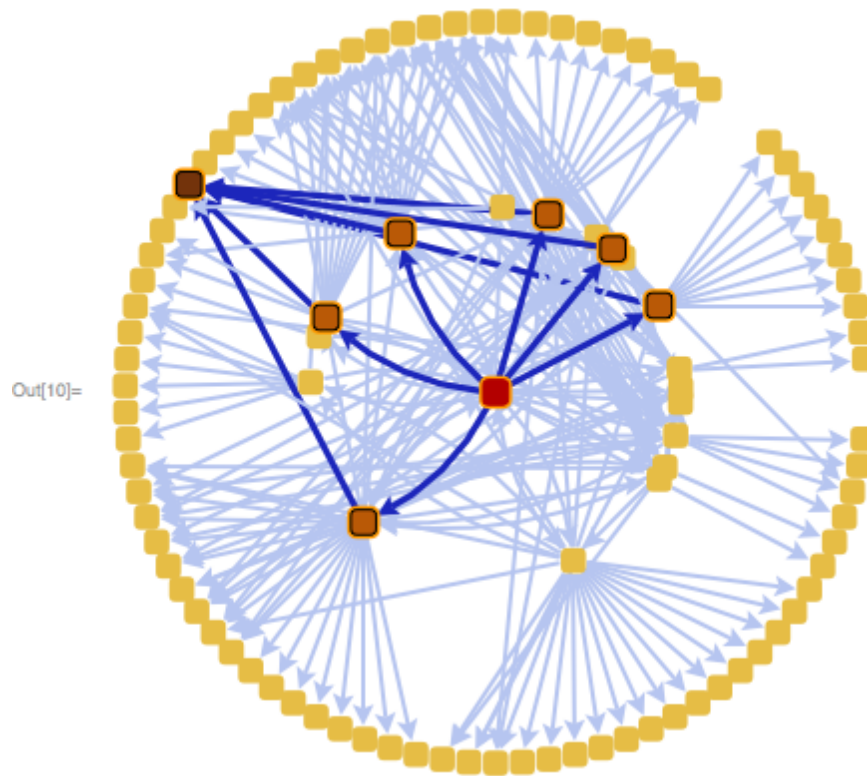
```

In[9]:= f = FindMinimumCostFlow[ag, amygdala (anatomical structure) ,
spinal cord (anatomical structure) , "OptimumFlowData"]

```

Out[9]= OptimumFlowData[ Flow value: 0]

```
In[10]:= f["FlowGraph"]
```



It is also noteworthy that, in addition to the brain's connectivity in the central nervous system, we have peripheral innervation integrated in our [AnatomyData](#) function. The motor commands from the spinal cord eventually reach the periphery.

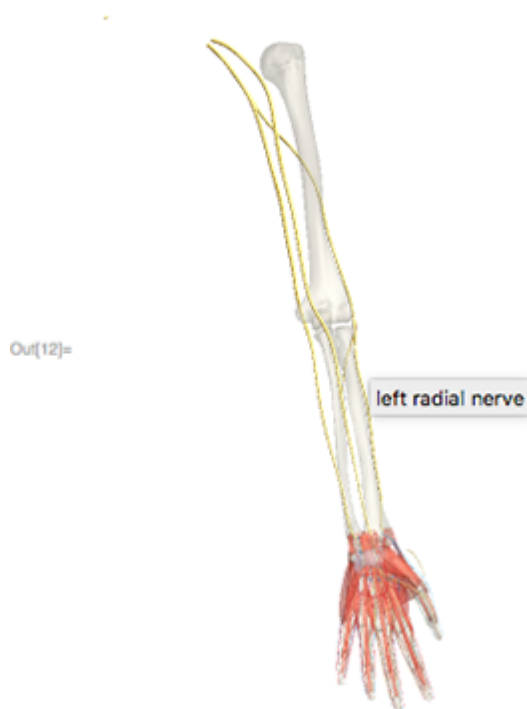
Find nerves that innervate the left hand:

```
In[11]:= nerve = EntityValue[left hand (anatomical structure), "NerveSupply"]
```

```
Out[11]= { left median nerve, left radial nerve, left ulnar nerve }
```

And visualize them in 3D with the [AnatomyPlot3D](#) function:

```
In[12]:= AnatomyPlot3D[{nerve, Opacity[.3], skeleton of left free upper limb (anatomical structure),  
left hand (anatomical structure)}] /. Annotation[x_, y_] => Tooltip[x, "Name" /. y]
```



Neuron Characteristics

We have looked at macroscopic pictures of our nervous system so far. Now let's look at the brain's functional unit, the neuron. Of course, we cannot characterize all the billions of neurons, but key features of a few hundred types of neurons are very similar across various mammalian species; these will be considered in detail.

A variety of properties are available for the "Neuron" entity type to describe physical, electrophysiological and spatial characteristics of individual types of neurons:

```
In[13]:= Entity["Neuron"]["SampleEntities"]
```

Out[13]= { cerebellum basket cell , cerebellum granule cell ,
dentate gyrus basket cell , hippocampus CA1 pyramidal cell ,
hippocampus CA3 pyramidal cell , inferior olivary nucleus principal neuron ,
neocortex basket cell , olfactory bulb (main) granule cell ,
retina bipolar cell , spinal cord ventral horn motor neuron }

```
In[14]:= Entity["Neuron"]["Properties"]
Out[14]:= { abbreviation , average AHP amplitude , average AHP duration ,
           average capacitance , average input resistance , average membrane time constant ,
           average resting membrane potential , average rheobase , average spike amplitude ,
           average spike half-width , average spike threshold , average spontaneous firing rate ,
           axon length , axon myelination , axon origin , axon projection laterality ,
           branching type , cell soma diameter , cell soma shape , cell soma size ,
           cellular synaptic target , defining criteria , dendrite location ,
           description , firing patterns , location of distant axon arborization ,
           location of local axon arborization , name , neurotransmitter released ,
           number of primary dendrites , role , soma location , spine density on dendrites }
```

We can get information on the types of neurons found in a particular brain region. For example, we can get a listing of neurons in the hippocampus, which is associated with emotional states, conversion of short-term to long-term memories and forming spatial memory:

```
In[15]:= EntityValue[Entity["AnatomicalStructure", "Hippocampus"], "Neurons"]
Out[15]:= { hippocampus CA1 IS-I neuron , hippocampus CA1 IS-II neuron ,
           hippocampus CA1 IS-III neuron , hippocampus CA1 LM(R) PP neuron ,
           hippocampus CA1 basket cell , hippocampus CA1 neurogliaform neuron ,
           hippocampus CA1 oriens lacunosum moleculare neuron , hippocampus CA1 pyramidal cell ,
           hippocampus CA2 basket cell broad , hippocampus CA2 pyramidal cell ,
           hippocampus CA3 IS-I cell , hippocampus CA3 IS-II cell ,
           hippocampus CA3 pyramidal cell , hippocampus CA1 ivy neuron ,
           hippocampus CA1 stratum oriens neuron , hippocampus CA1 trilaminar neuron ,
           hippocampus CA2 basket cell narrow , hippocampus CA2 bistratified cell broad ,
           hippocampus CA2 bistratified cell narrow , hippocampus CA3 axo-axonic cell ,
           hippocampus CA3 basket cell , hippocampus CA3 lacunosum moleculare neuron ,
           hippocampus CA3 oriens lacunosum moleculare neuron ,
           hippocampus CA3 radiatum neuron ,
           hippocampus CA3 spiny CR cell , hippocampus (CA) pyramidal cell }
```

Collecting further details, list the set of neurons whose axons arborize at the CA1 alveus area of the hippocampus:

```

In[16]:= EntityValue[
  EntityClass["Neuron",
    {"LocationOfDistantAxonArborization" → CA1 alveus (anatomical structure) }],
  "Entities"]

Out[16]= { hippocampus CA1 IS-III neuron , hippocampus CA1 IS-I neuron ,
  hippocampus CA1 ivy neuron , hippocampus CA1 lacunosum moleculare neuron ,
  hippocampus CA1 stratum oriens neuron ,
  hippocampus CA1 trilaminar neuron , hippocampus CA2 bistratified cell broad ,
  hippocampus CA2 bistratified cell narrow , hippocampus CA2 pyramidal cell }

```

Neurons transmit electrical signals to communicate with one another. Physical characteristics and patterns of their spikes, known as action potentials, differ across different neuron types.

We can obtain experimentally measured electrophysiological properties of hippocampus CA1 pyramidal cells:

```

In[17]:= Dataset@
  EntityValue[ hippocampus CA1 pyramidal cell (neuron) , "NonMissingPropertyAssociation"][[
    ;; 10]]

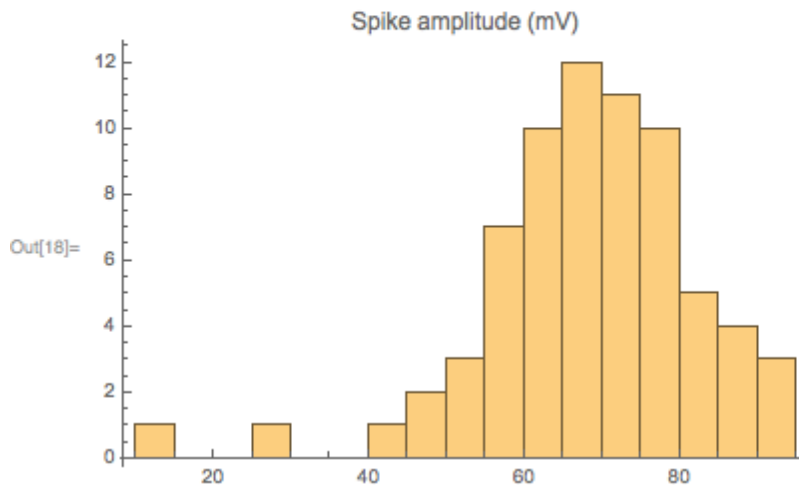
Out[17]=

```

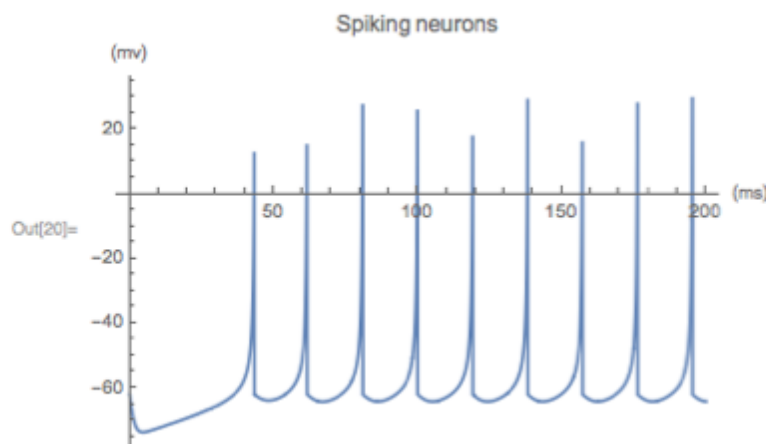
average AHP amplitude	7.0 mV
average AHP duration	19. ms
average capacitance	81. pF
average input resistance	11. MΩ
average membrane time constant	22. ms
average resting membrane potential	-64. mV
average rheobase	28. pA
average spike amplitude	87. mV
average spike half-width	1.3 ms
average spike threshold	-47. mV

Here we can visually recognize how spike characteristics vary across different neuron types:

```
In[18]:= Histogram[EntityValue[EntityList["Neuron"], average spike amplitude,
    "NonMissingEntityAssociation"], 20, PlotLabel -> "Spike amplitude (mV)"]
```



A single neuron's spike propagation can be simulated with the well-known Hodgkin and Huxley model ([A. L. Hodgkin and A. F. Huxley, 1952](#)) based on four differential equations involving voltages and currents. Also, there are biologically realistic computational models accommodating Hodgkin and Huxley's concepts developed to simulate ensembles of spikes in a population of neurons ([E. M. Izhikevich, 2004](#)). We can better understand how neurons excite/suppress one another to transmit information by modeling the neurons' electrical spikes and comparing their patterns of activities with experimentally measured ones:



Cognitive Map

After looking at microscopic features in our brain, let us finally explore the brain's macro-scale executive function. Thanks to recent advances in imaging techniques to visualize brain activity in various cognitive states, we can map out cortical areas that are associated with specific cognitive processes. Brain areas associated with specific functions such as memory, decision making, language, emotional state, visual perception, etc. are well characterized with the appropriate activity-based fMRI analysis.

Using the `EntityValue` query with the `AnatomicalFunctionalConcept` entity type, we can find more information on hierarchically categorized brain activities:

```
In[21]: Dataset@EntityValue[decision making (anatomical functional concept),
    {definition, associated anatomical sites, broader concepts, narrower concepts,
    superset concepts, subset concepts, standard experimental tasks},
    "PropertyAssociation"]
```

Out[21]=

Definition	the deliberate selection of a single alternative amongst multiple alternatives
Associated Anatomical Sites	{ frontal lobe , striatum , prefrontal cortex , striatum of right cerebral hemisphere , Brodmann area 47 , orbitofrontal cortex , Brodmann area 9 }
Broader Concepts	---
Narrower Concepts	{ animacy decision , categorization , emotional decision making , feature comparison , inference , intertemporal choice , reasoning }
Superset Concepts	{ set shifting }
Subset Concepts	{ decision certainty , reinforcement learning , response selection , risk , uncertainty , working memory }
Standard Experimental Tasks	{ Cambridge risk task , hungry donkey task , information sampling task , Iowa gambling task }

Here we can look up the categories of functions associated with each cerebral lobe and create a simple cortical map:

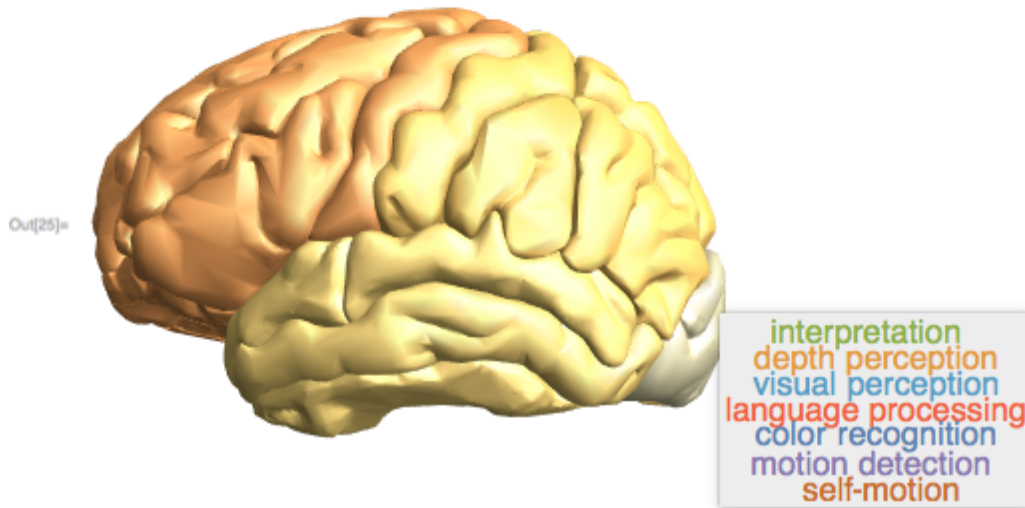
```

In[22]:= lobe = { frontal lobe (anatomical structure), insula (anatomical structure),
                 parietal lobe (anatomical structure), temporal lobe (anatomical structure),
                 limbic lobe (anatomical structure), occipital lobe (anatomical structure) };

lobeToConcept =
  # → EntityValue[EntityClass["AnatomicalFunctionalConcept",
    { associated anatomical sites → #}], "Entities"] & /@ lobe;

ia = Normal@EntityValue[lobe, "ImageAtoms", "EntityAssociation"];
AnatomyPlot3D[Riffle[Table[ColorData["BeachColors"][i], {i, 0.1, .6, .1}], lobe],
  ImageSize → 400, ViewPoint → Right] /.
  Annotation[a_, b_] →
  Tooltip[a,
    Which @@
      (Sequence @@@
        ({MemberQ[#, /. ia, "Entity" /. b], WordCloud[#, /. lobeToConcept, ImageSize → 150]} & /@
          lobe))]

```



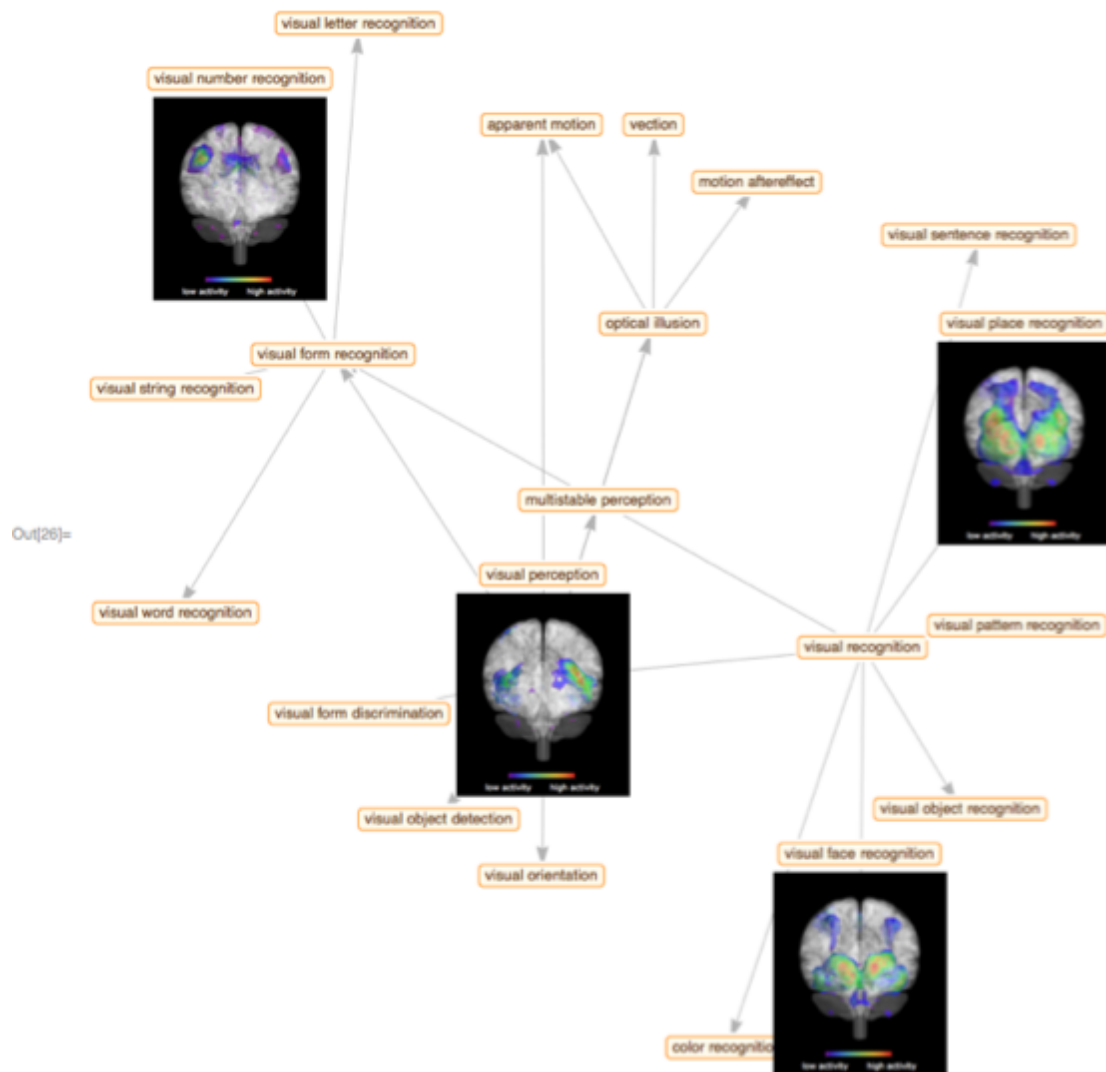
We are not limited to the abstract representation of cortical maps; fMRI-based statistical maps of brain activity are also available.

Let's look at how we perceive the visual world. A key aspect of visual perception is the subprocess (concept) of cognition as our brain categorizes our visually perceived faces, places, words, numbers, etc. with distinctive patterns of activity. The following graph illustrates how these concepts are hierarchically organized. Some areas of brain activation are highlighted (brain images are seen from the rear):

```

In[26]> g = NestGraph[Cases[EntityValue[#, "NarrowerConcepts"], _Entity] &,
  visual perception (anatomical functional concept), 2];
act = EntityValue[VertexList[g], "BrainGraphicBack"];
Graph[EdgeList[g],
  VertexLabels ->
  MapThread[
    #1 -> If[MemberQ[{
      visual perception (anatomical functional concept),
      visual face recognition (anatomical functional concept),
      visual number recognition (anatomical functional concept),
      visual place recognition (anatomical functional concept)
    }, #1],
    Placed[Grid[{{Rasterize[#1]}, {#2}}], Center],
    Placed[Rasterize[#1, ImageSize -> {Automatic, 26}], Center] &, {VertexList[g], act}],
  VertexSize -> Small, GraphLayout -> "RadialEmbedding", PlotTheme -> "Monochrome",
  AspectRatio -> 1.2, ImageSize -> 900, ImagePadding -> 60] // Image

```



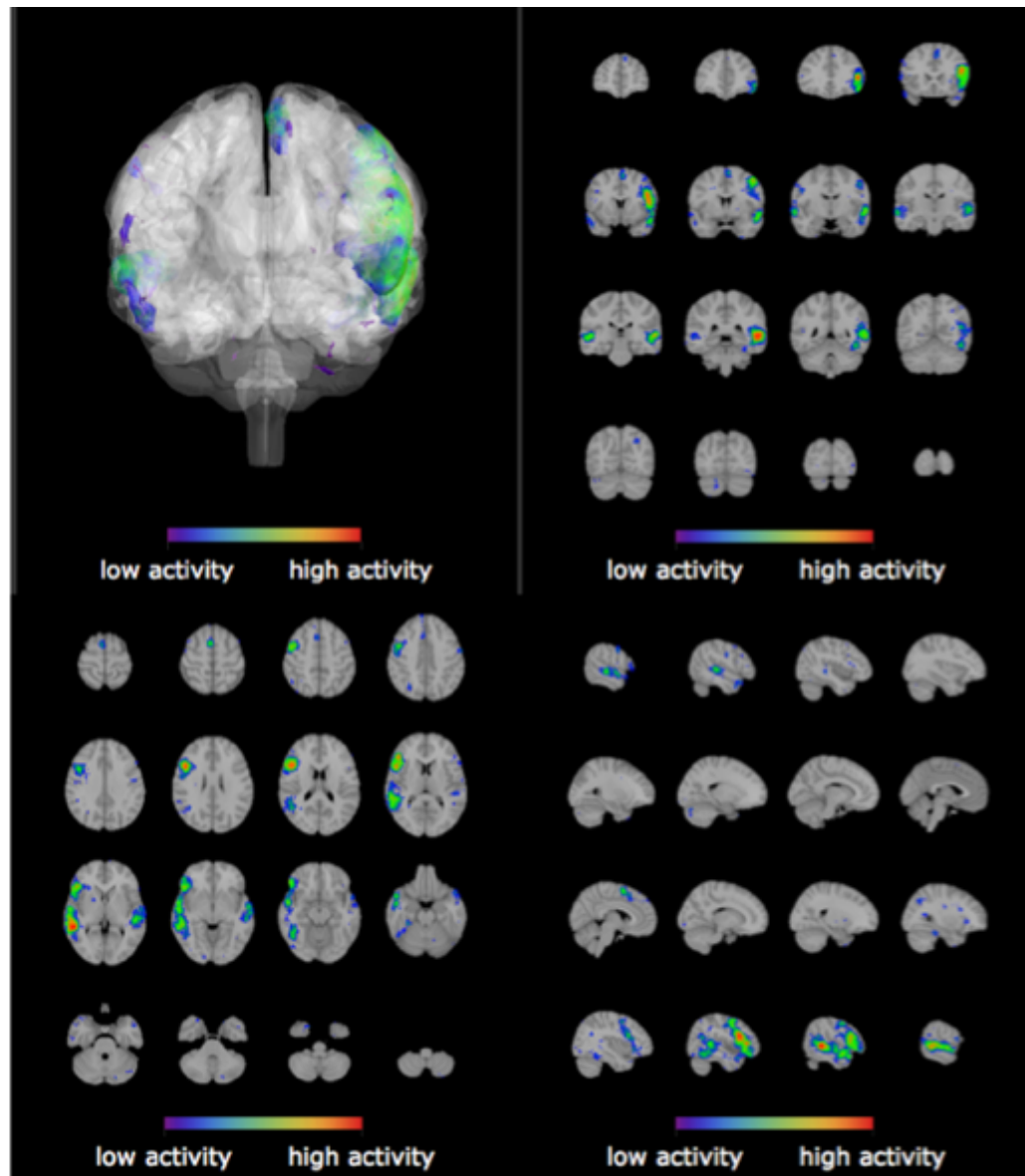
OK, let's look further. Visually perceived words, sentences, faces, etc., in turn, affect “language” and “emotion”:

```

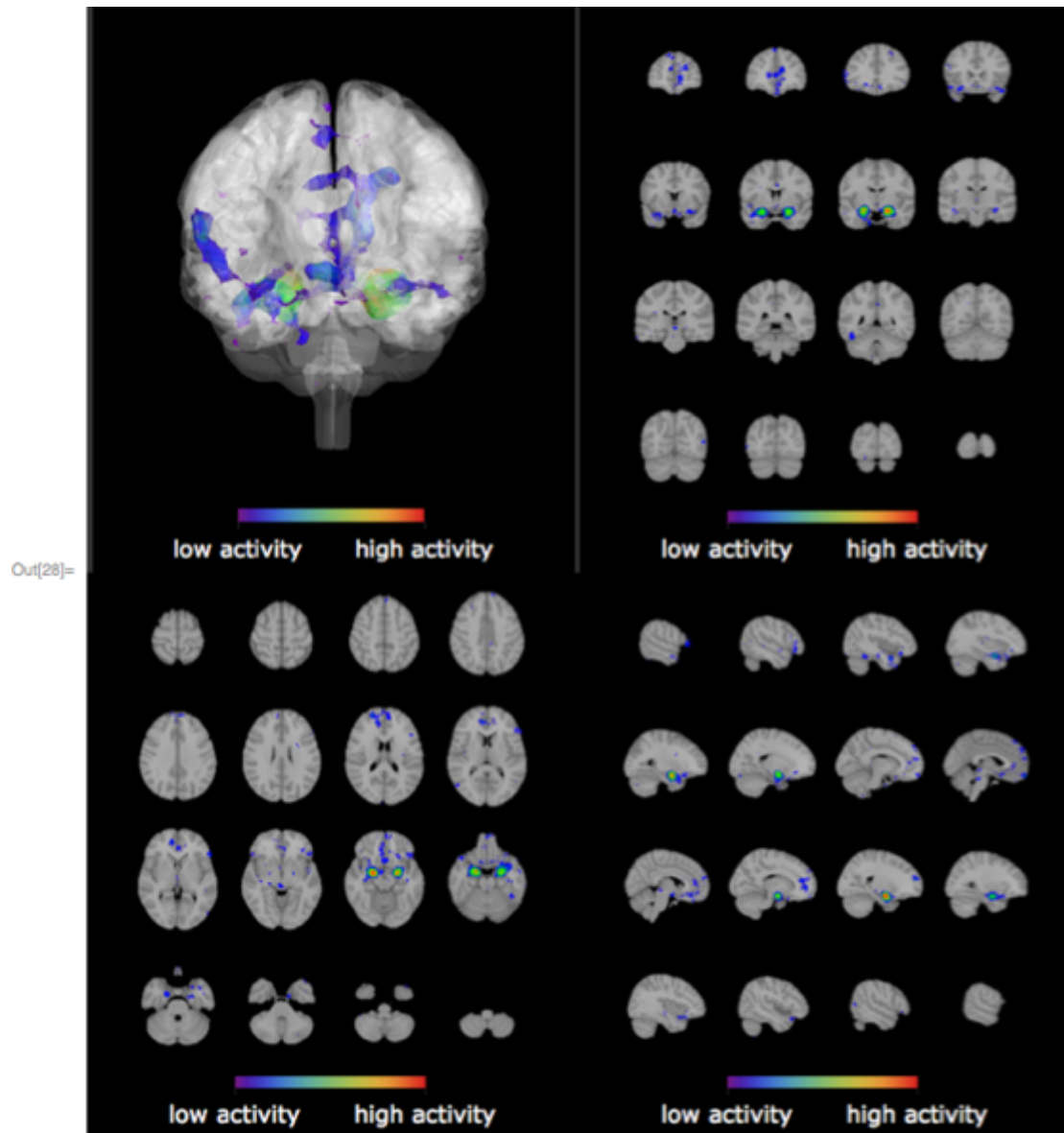
In[27]:= ImageCollage@EntityValue[ language (anatomical functional concept) ,
  { brain graphic front , brain image coronal slices , brain image horizontal slices ,
    brain image sagittal slices } ]

```

Out[27]=



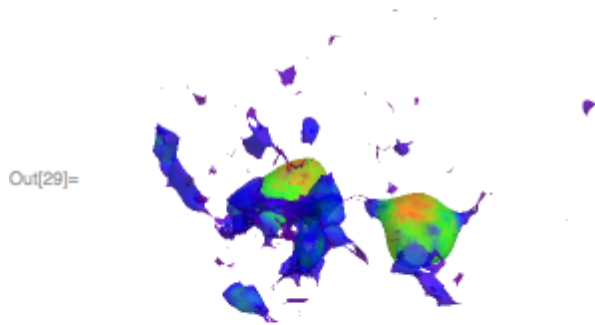
```
In[28]:= ImageCollage@EntityValue[emotion (anatomical functional concept),
  {brain graphic front, brain image coronal slices, brain image horizontal slices,
    brain image sagittal slices}]
```



We can confirm that the amygdala (remember, the left and right amygdalae found near the center of the brain) is actively involved in emotions. If you want to learn more about these individual models, they are also available in 3D polygon data and ready to be aligned to our 3D brain model in `AnatomyData` for further computation.

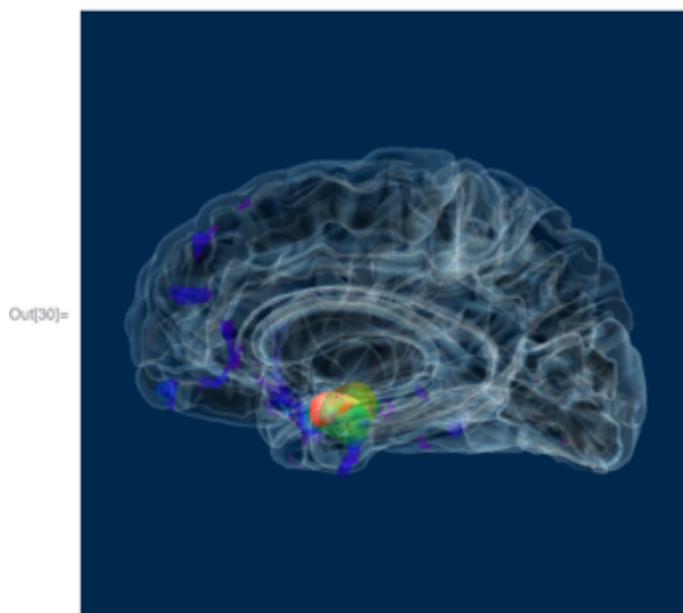
Here is the brain activation area 3D graphic associated with emotion:

```
In[29]:= em = emotion (anatomical functional concept) ["ActivationAreas3DGraphic"]
```



We can combine that graphic together with the brain model for visual comparison (the amygdala is highlighted in red; the right cerebral hemisphere is shown here for demonstration):

```
In[30]:= AnatomyPlot3D[{em /. Opacity[_] => Opacity[.4],
  {Directive[Specularity[White, 30], Hue[.58, 0, 1, .12], Lighting -> light],
    right cerebral hemisphere (anatomical structure) },
  {Glow[Red], right amygdala (anatomical structure) }},
PlotRange -> right cerebral hemisphere (anatomical structure) , Background -> Hue[.58, 1, .3],
ViewPoint -> Right, SphericalRegion -> True,
BaseStyle -> {RenderingOptions -> {"DepthPeelingLayers" -> 20}}]
```



It's fascinating to learn how our brain is organized and how it coordinates the processes in our nervous system. As we know, there is still a lot to be learned about human cognition, and exciting discoveries are being made every day. As we gain additional insights, we continue to expand our knowledgebase to attain a better and deeper understanding of the human nervous system.

Stay tuned for more neuroscience content to come!

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